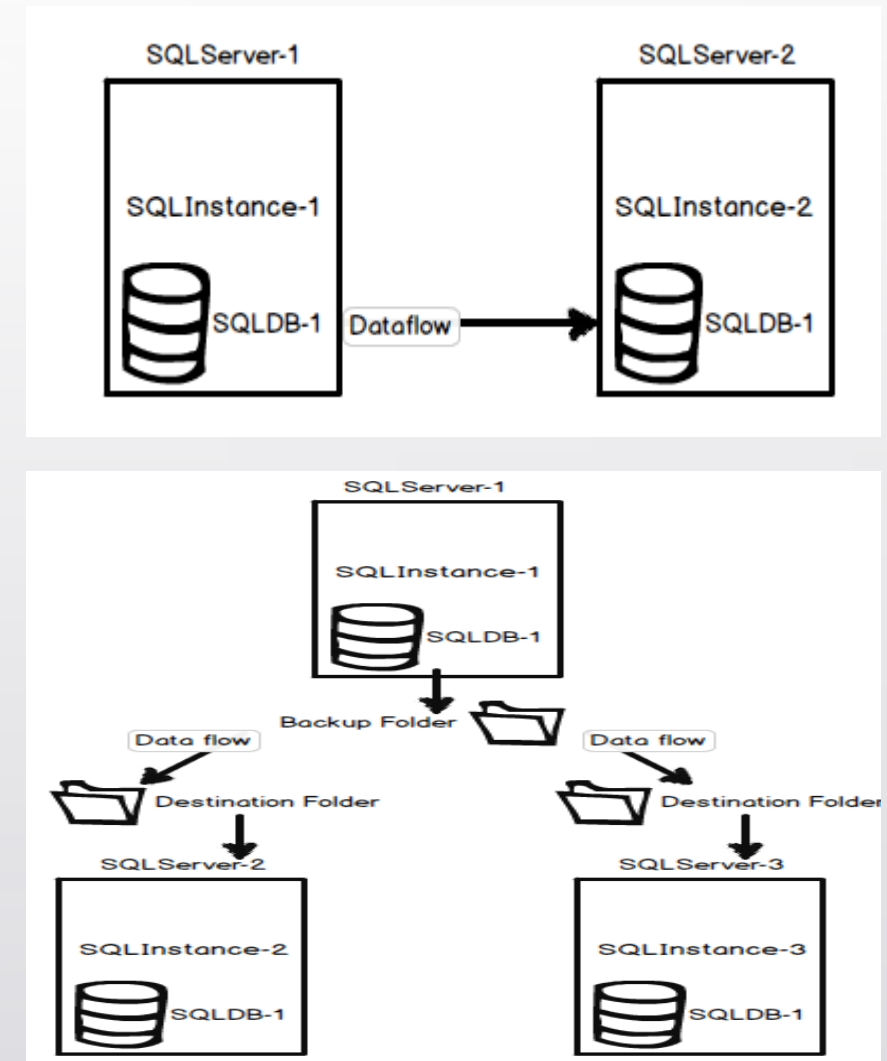




Log Shipping and Configuring Log Shipping

SQL Server Log Shipping

- SQL Server log shipping is a technique which involves two or more SQL Server instances and copying of a transaction log file from one SQL Server instance to another.
- The process of transferring the transaction log files and restoring is automated across the SQL Servers.
- As the process result there are two copies of the data on two separate locations.
- A log shipping session involves the following steps:
 - Backing up the transaction log file on the primary SQL Server instance
 - Copying the transaction log backup file across the network to one or more secondary SQL Server instances
 - Restoring the transaction log backup file on the secondary SQL Server instances





SQL Server Log Shipping Operating Modes

- There are two available modes and they are related to the state in which the secondary, log shipped, SQL Server database will be:
- Standby mode – the database is available for querying and users can access it, but in read-only mode
 - The database is not available only while the restore process is running
 - Users can be forced to disconnect when the restore job commence
 - The restore job can be delayed until all users disconnect themselves
- Restore mode – the database is not accessible



Log Shipping Pros and Cons

- it's reliable and tested in details.
- it's relatively easy to set up and maintain.
- there is a possibility for failover between SQL Servers, data can be copied on more than one location etc.
- Log shipping can be combined with other disaster recovery options such as AlwaysOn Availability Groups, database mirroring, and database replication.
- SQL Server log shipping has low cost in human and server resources
- The main disadvantages in the SQL Server log shipping technique are:
 - Need to manage all the databases separately, there isn't possibility for an automatic failover,
 - Secondary database isn't fully readable while the restore process is running



Setting up Log Shipping

- The database log shipping setup needs to be initiated from the principal server using the SQL Server Management Studio wizard. The first step defines transaction log backup settings:
- A network path to the backup
- How long backup files should be kept before deleting
- An alert if no backup is taken
- The backup job itself
 - Schedule of the job
 - Schedule type
 - Frequency
 - Duration



Setting up Log Shipping (contd...)

- The next step defines secondary databases which involve choosing the secondary SQL Server instance and secondary database. The full database backup, from the primary database, must be restored on the secondary server before log shipping commences.
- After initializing the secondary database you must define the copy folder where the transaction log backups from the primary server will be stored.
- The final step involves choosing from two available modes: The No recovery – Restore mode and Standby mode. You can also delay the restoring process and set up an alert if no restore occurs within the specified time.
- Once the log shipping is ready for use, it will run in the background, and if the problem occurs the alert will signalize the problem.